

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [103, 99, 94, 79, 95, 95, 98, 93, 91, 85]

Tap amplitudes (pixels): [173.07223925286226, 173.08090593707902, 171.73234989366446, 165.0272704737008, 146.37281168304446, 165.71059109181888, 164.20109622045769, 163.783393541592, 162.69296235547498, 150.3761949245957, 151.7003625572464]

GPT-5

MDS-UPDRS Item 3.4: Finger Tapping – Automated Feature-Based Assessment

Given features

- Number of taps: 11 (i.e., 10 intervals listed; appears the subject performed 11 taps, which is fine since instruction targets 10 taps)
- Tap periods (frames): [103, 99, 94, 79, 95, 95, 98, 93, 91, 85]
 - Trend: modest speeding early (103 → 79), then relatively stable 79–99 range; no obvious prolonged arrests/freezes; no >5 interruptions apparent from timing alone.
- Tap amplitudes (pixels): [173.07, 173.08, 171.73, 165.03, 146.37, 165.71, 164.20, 163.78, 162.69, 150.38, 151.70]
 - Pattern: near-constant for first 2 taps (~173), slight drop by tap 3 (~172), more marked drop by tap 4 (~165), pronounced dip at tap 5 (~146), then partial recovery to mid-160s, followed by a gradual decline toward ~150 by taps 10–11.

Interpretation vs. scoring anchors

- Interruptions/hesitations/halts:
 - Periods show variability within a reasonable band (79–103 frames) without evidence of long arrest. No data indicating >5 interruptions; no explicit hesitations flagged. This argues against score 3a or 4 on interruption grounds.
- Speed:
 - No clear moderate slowing trend; intervals actually shorten early and remain relatively similar thereafter. This does not meet “moderate slowing.”
- Amplitude decrement:
 - The amplitude is stable for the first 2 taps and begins to decrease by tap 3–4, with a clear dip by tap 5, some recovery, then a gradual decline through taps 9–11.
 - Scoring anchors:
 - 1c: decrement near the end.
 - 2c: decrement midway through the 10 taps.
 - 3c: decrement starting after the 1st tap.
 - Here, the decrement becomes evident by taps 3–5, i.e., not just “near the end,” and not “starting immediately after the 1st tap.” This best aligns with “midway” decrement (score 2c).

Given the lack of clear frequent interruptions or moderate slowing, and the presence of a midway amplitude decrement pattern, the amplitude criterion governs the score.

Score: 2
